

| <div> <div>Science.7.10A/B</div> <div> <div>Biome Investigation</div> </div> </div>                                                                                                         |                                                                                                                                                                           |                                                                                                                                                                                                                       | Name _____<br><br>Date _____ |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| <b>Essential Questions</b><br>1. How do different environments support different varieties of organisms?<br>2. How does this biodiversity contribute to the sustainability of an ecosystem? | <b>Vocabulary</b><br>Biodiversity<br>Biomes<br>Microhabitat<br>Species<br>Climate<br><b>Scientific Investigation and Reasoning Skills</b><br>7.1B, 7.2A, 7.2C, 7.2D, 7.3A | <b>Objectives</b><br>1. Describe the characteristics of different biomes and the animals and plants found in the different biomes.<br>2. Describe how biodiversity contributes to the sustainability of an ecosystem. |                              |

**Problem:** Observe and describe characteristics of the following types of biomes:

- |                               |                       |
|-------------------------------|-----------------------|
| 1. Tropical Rainforest        | 6. Grassland          |
| 2. Tundra                     | 7. Freshwater Aquatic |
| 3. Taiga (Coniferous Forest)  | 8. Marine Aquatic     |
| 4. Desert                     |                       |
| 5. Temperate Deciduous Forest |                       |

**Background information:** Organisms are classified into Domains and Kingdoms, and those organisms within these taxonomic groups share similar characteristics. These characteristics allow them to interact with the living and nonliving parts of their ecosystem. Biomes are regions of the world that share similar plant structures, animals, climate and weather. There are generally considered to be eight major biome categories (listed above). Six are terrestrial (on land) and two are aquatic (in the water).

**Materials:**

Science Journal

Computer with Internet Access

**Procedure:**

1. Log on to your computer, and go to <http://www.mbgnet.net/index.html>
2. Create a Data Table to record the following for each of the 8 biomes:
 

|                          |                      |
|--------------------------|----------------------|
| a. Location              | e. Animal species    |
| b. Average temperature   | f. Interesting facts |
| c. Average precipitation |                      |
| d. Plant species         |                      |

Your data table might look like this one, but with MUCH MORE SPACE to record your findings. Consider allocating 2 page for this table, with about 4 lines per Biome:

| Characteristics     | Location | Average Temperature | Average Precipitation | Plant Species | Animal Species | Interesting Facts |
|---------------------|----------|---------------------|-----------------------|---------------|----------------|-------------------|
| <b>Biomes</b>       |          |                     |                       |               |                |                   |
| Tropical Rainforest |          |                     |                       |               |                |                   |
| Tundra              |          |                     |                       |               |                |                   |

|                                     |  |  |  |  |  |  |
|-------------------------------------|--|--|--|--|--|--|
| <i>Taiga (Coniferous) Forest</i>    |  |  |  |  |  |  |
| <i>Desert</i>                       |  |  |  |  |  |  |
| <i>Deciduous (Temperate) Forest</i> |  |  |  |  |  |  |
| <i>Grassland</i>                    |  |  |  |  |  |  |
| <i>Freshwater Aquatic</i>           |  |  |  |  |  |  |
| <i>Marine Aquatic</i>               |  |  |  |  |  |  |

- Investigate each Biome on the website, and record your findings in your data table.

### **Conclude and Apply**

*Review your data. Answer the following questions in your science notebook using complete sentences.*

- You are blindfolded and dropped by helicopter into an unknown environment. When you remove your blindfold, you must determine what Biome you are standing in. Describe what process you would use to determine the Biome you are in.
- What are the similar characteristics of the animals within each of these 8 biomes?
- What are the similar characteristics of the plants within each of these 8 biomes?
- Would it be possible for a moose, who primarily lives in a taiga biome, to be found in a desert biome? Why? Compare the similarities and differences of the two biomes in your answer.
- How do the different biomes support different types of organisms?
- Pick a biome, and then create an organism that would live in that biome. What are the characteristics of this organism that make it ideal for each of the six characteristics of that biome?